EXPERIMENT 1: COMPONENT CODING AND MULTIMETER RESISTOR MEASUREMENT

OBJECTIVE:

1. Understand how to set digital multimeter for resistance measurement
2. To acquaint students with the skill to read resistor values based on color code
3. Measure resistor in series and parallel connection.
4. Calculate measurement deviation percentage

INTRODUCTION:

Resistor coding

The color code technique is used to show resistance values of carbon resistors without having to measure it. In this technique color bands are printed on the resistor. The procedure for determining the resistance of a color-coded resistance is described in Table 1.

Table 1.1: Resistor color coding

Color	**1st Band (1st Significant Digit)**	**2nd Band (2nd Significant Digit)**	**3rd Band (Multiplier)**	**4th Band (Tolerance)**
Black	0	0	1	
Brown	1	1	10^1	1%
Red	2	2	10^2	2%
Orange	3	3	$1\ 0^3$	3%
Yellow	4	4	10^4	4%
Green	5	5	10^5	
Blue	6	6	10^6	
Violet	7	7	10^7	
Grey	8	8	10^8	
White	9	9	10^9	
Gold	-	-	0.1	5%
Silver	-	-	0.01	10%
No Color	-	-	-	20%

The first two bands determine the first two digits of the resistor value, while the third band

determines the power of 10-multiplier. For the resistor with value less than 10 Ω the third band is either silver or gold. The forth band is the percent tolerance for the chosen resistor. If resistors have only three bands, it means the forth band has no color. Sometimes a fifth band is employed for some high precision resistor where the first three bands represent the significant digit. The forth band is the multiplier while the fifth band is the tolerance. In the other case, for some standard 4-band code, a fifth band may indicate the manufacturer's special code for some physical characteristic or failure rate of the component.

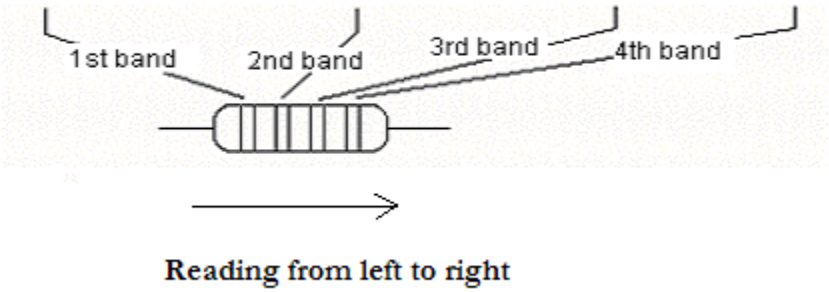
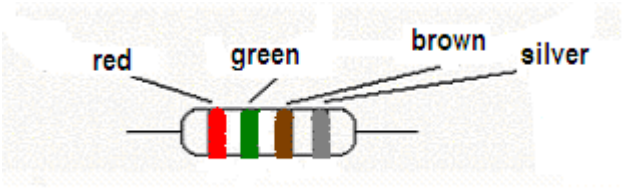


Figure 1.1: Reading resistor color coding

Example 1:



The value of this resistor is 25 x 10^1± 10% = 250± 10% ohms

Minimum value – 225 Ω
Maximum value – 275 Ω

Resistance should never be measured in a live network due to the possibility of damaging the

meter with excessively high currents and obtaining readings that have no meaning. In a constructed circuit, to measure a single resistance value, just take off one end of its terminal to avoid the effect of other resistances in the circuit. This applies in the same manner to the other components such as capacitor and inductor.

Breadboard

When building a "permanent circuit" the components can be "grown" together (as in an integrated circuit), soldered together (as on a printed circuit board), or held together by screws and clamps (as in house wiring). In lab, we want something that is easy to assemble and easy to change. We also want something that can be used with the same components that "real" circuits use. Most of these components have pieces of wire or metal tabs sticking out of them to form their terminals.

A breadboard is used to make up temporary circuits for testing or to try out an idea. No soldering is required so it is easy to change connections and replace components. Parts will not be damaged so they will be available to re-use afterwards.

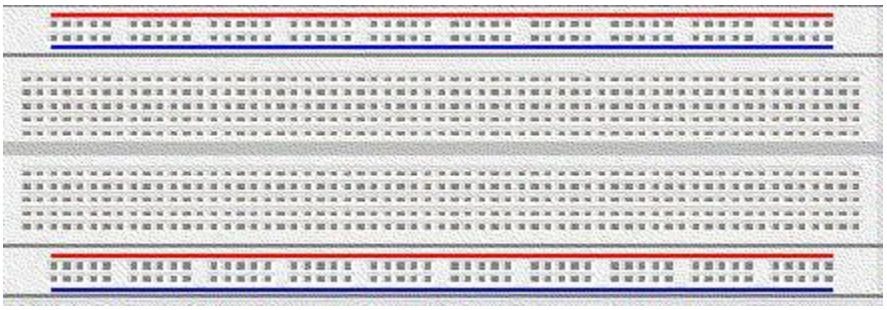


Figure 1.3: Front look of a typical small breadboard used in the laboratory

The breadboard has many strips of metal (usually copper) which run underneath the board.

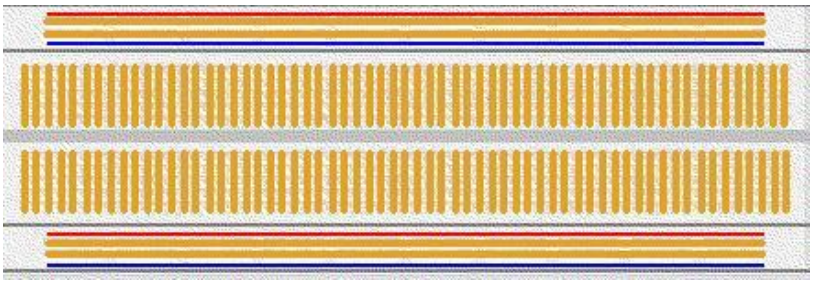


Figure 1.4: The metal strips layout

Multimeter

Multimeter is a basic tool in electric and electronic fields. It is a multipurpose device to measure voltage, current and resistance. Basically there are two types of multimeter used either in the education or industrial field based on the electronic circuits inside them: analog and digital meters.

The analog meter, broadly known as VOM (volt-ohm-miliammeters) uses a mechanical moving pointer which indicates the measured quantity on a calibrated scale. It requires the user a little practice to interpret the location of the pointer.

The digital meter broadly known as DMM (digital multimeter) used number or numerical display to represent the measured quantity. It has high degree of accuracy and can eliminate usual reading errors compared to the analog meters. Students should be adept at using both meters throughout their studies.

The connection of the multimeter to measure different electrical quantities is shown in both schematic diagram and real wiring illustration in the laboratory in Figure 1.5.

Resistor in series and parallel

A series circuit is a circuit in which resistors are arranged in a chain, so the current has only one path to take. The current is the same through each resistor. The total resistance of the circuit is found by simply adding up the resistance values of the individual resistors:

Total resistance of resistors in series:

$$RT = R1 + R2 + R3 + ... \quad (1)$$

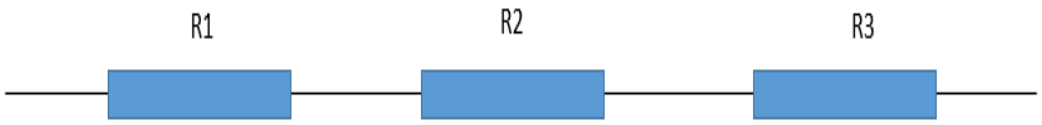


Figure 1.6: Resistor connected in series

A parallel circuit is a circuit in which the resistors are arranged with their heads connected together, and their tails connected together. The current in a parallel circuit breaks up, with some flowing along each parallel branch and re-combining when the branches meet again. The voltage across each resistor in parallel is the same.

The total resistance of a set of resistors in parallel is found by adding up the reciprocals of the resistance values, and then taking the reciprocal of the total:

Total resistance of resistors in parallel:

$$1 / R = 1 / R1 + 1 / R2 + 1 / R3 +... \quad (2)$$

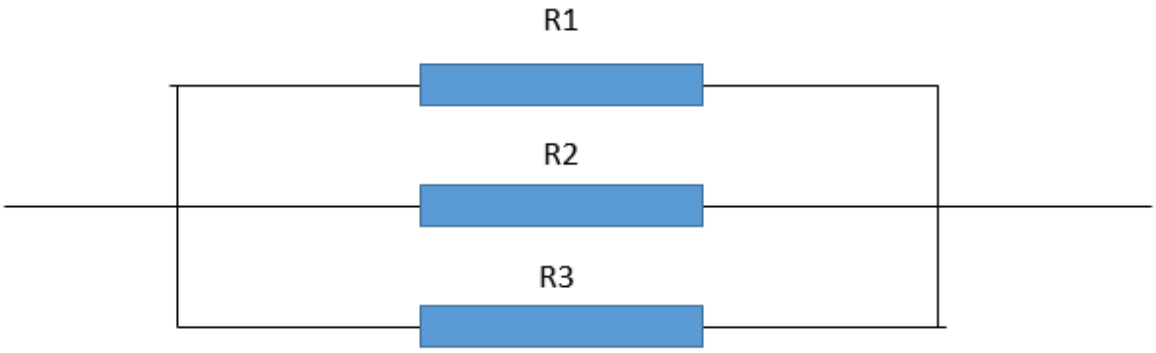


Figure 1.7: Resistor connected in parallel

Deviation percentage calculation procedure.

Deviation = Measured value – label value

Deviation % = (Deviation / label value) x 100

EQUIPMENT AND COMPONENT

1. HMC8012
2. resistor 10ohm, 100ohm, 1k ohm, 10k ohm and 100k ohm
3. Breadboard
4. Test lead

PROCEDURE

A. USING MULTIMETER TO MEASURE RESISTOR

1. Setup the instrument in a slightly inclined position, fold out the small hinges on the bottom stands ii. Make sure that the stands are completely folded out to ensure a stable position.



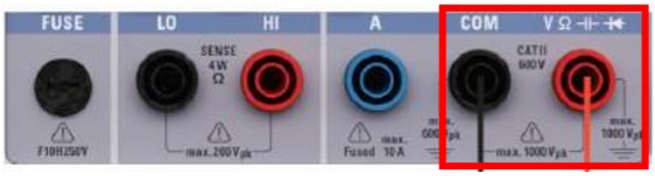
2. Turn ON/OFF the main switch button on the rear panel



3. Turn ON/OFF the power switch on the front panel. The LED of power switch will shows green for the ON indicator and red for the OFF indicator.



4. Connect test lead to V and COM connectors on bottom right



5. Press the Ω soft key to start measure the resistance



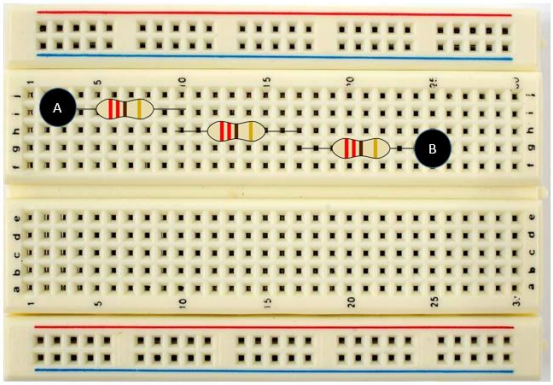
6. Connect both test lead with a resistor (10ohm).



7. Measure resistance of resistor. Record the measured value in Table 1.3
8. Then determine the measurement deviation percentage.

B: MEASURE RESISTOR IN SERIES CONNECTION

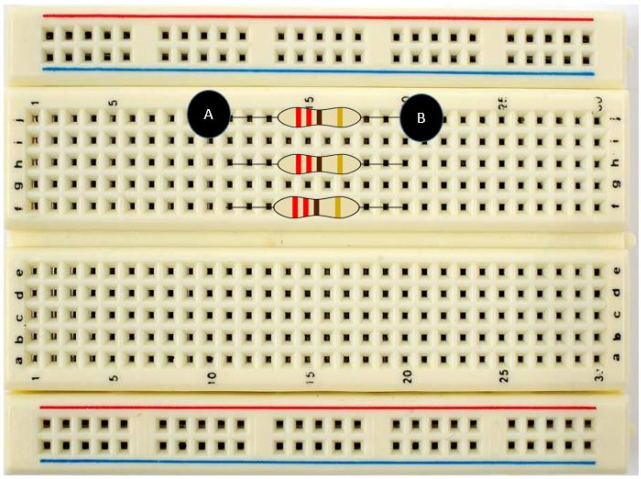
1. Connect resistor 10 Ω, 10k Ω and 100k Ω in series as per below diagram and calculate the resistance value by using equation 1.



2. Probe resistor at point A and B to measure total resistance. Record the total resistance value Table 1.4.
3. Then determine the measurement deviation percentage.

C: MEASURE RESISTOR IN PARALLEL CONNECTION

1. Connect resistor 10 Ω, 10k Ω and 100k Ω in parallel as per below diagram and calculate the resistance value by using equation 2.



2. Probe resistor at point A and B to measure total resistance. Record the total resistance value in Table 1.5
3. Then determine the measurement deviation percentage

RESULT

READING RESISTOR BY COLOR CODING

Determine the nominal value or color bands of a particular resistor based on color coding technique for each case given in Table 1.2 below. Check your answers with the measured values in the laboratory.

Table 1.2: Exercise for determining resistor values by color coding and measurement

	COLOR BAND				
No.	Band 1	Band 2	Band 3	Band 4	Nominal Value
1.	orange	orange	black	gold	
2.	orange	white	red	gold	
3.	brown	green	orange	gold	
4.	red	violet	orange	gold	
5.	orange	white	orange	gold	
6.	brown	black	gold	no color	
7.	red	violet	orange	gold	

PART A: READING RESISTOR BY USING DIGITAL MULTIMETER

Table 1.3: Measured resistance using DMM

Resistor	**R value Ω**	**Measured value**	**Deviation**	**Deviation %**
1	10 ohm			
2	100 ohm			
3	1k ohm			
4	10k ohm			
5	100k ohm			

PART B: MEASURE RESISTOR IN SERIES CONNECTION

Table 1.4: Measured resistance using DMM

Resistor	R_T value **Ω**	Measured value	Deviation	Deviation %
Series				

PART C: MEASURE RESISTOR IN PARALLEL CONNECTION

Table 1.5: Measured resistance using DMM

Resistor	R_T value **Ω**	Measured value	Deviation	Deviation %
Parallel				

DISCUSSION

1. Elaborate your result and findings in this experiment.
2. Explain why tolerance and deviation is importance in instrument or component measurement.
3. Digital or alphabet code method can be used to read resistor value. Explain the digital/alphabet code method compared to color coding reading and measurement using DMM.